

University Admission Management Sysytem



Submitted by:

M.Sharjeel Sarwar 2025-CS-1205

Section C

Supervised by:

Sir.Talha Mubashir

Department of Computer Science

University of Engineering and Technology

Lahore Pakistan

Short Description :

This Project University Admission Management System (UAMS) is a console-based C++ program that allows an authenticated user to manage student admission records. The system supports user registration and login, and after successful authentication, enables adding, updating, deleting, viewing student records, and calculating merit. All student data is stored in files so that records persist even after the program is closed and restarted.

Users of Game :

There is admin and a User in this Application.

Functional requirements :

- The system allows the user to **register and log in**.
- The system allows the user to **add student records**.
- The system allows the user to **update student information**.
- The system allows the user to **delete student records**.
- The system allows the user to **view all student records**.
- The system calculates **student merit** based on marks.
- The system **saves and loads data from files**.

Wire Frames :

Welcome screen

```
#=====#
# University Admission Management System #
#=====#

1. Register User
2. Login
Choose option:
```

Login Screen For Admin

```
#####  
#===== UAMS LOGIN =====#  
#####  
  
Enter Username: Sharjeel  
Enter Password: 123456@.
```

Figure : Admin Login

Main Menu

```
#=====#  
# University Admission Management System #  
#=====#  
  
#=====#  
##<<<<< MENU >>>>>##  
#=====#  
  
1 - Add Students.  
2 - Update record.  
3 - Calculate Merit.  
4 - Delete Student.  
5 - View.  
6 - Exit.  
CHOOSE AN OPTION FROM MENU:
```

Complete Code:

```
#include<iostream>
#include<fstream>
#include<conio.h>
#include<iomanip>
#include<string>
using namespace std;

void Register();
bool login();
void Header();
void Menu();

int loadFromFile(string name[], string Cnic[], float Matric[], float Fsc[], float
ECAT[], float Merit[]);

void saveToFile(string name[], string Cnic[], float Matric[], float Fsc[], float
ECAT[], float Merit[], int indx);

int addstudent(string name[],string Cnic[],float Matric[],float Fsc[],float ECAT[] ,
float Merit[], int num,int indx);

int update(string name[], string Cnic[], float Matric[], float Fsc[], float ECAT[],
float Merit[], int indx);

void View(string name[], string Cnic[], float Matric[], float Fsc[], float ECAT[],
float Merit[], int &indx);

void merit(string name[], string Cnic[], float Matric[], float Fsc[], float ECAT[],
float Merit[], int indx);

void Delete(string name[], string Cnic[], float Matric[], float Fsc[], float ECAT[],
float Merit[], int &indx);

string getfield(string record, int field);

void readData(string names[], string passwords[], int &indx);
```

```
int isPresent(string name, string password, string names[], string passwords[], int  
idx);
```

```
string getfield(string record, int field)  
{  
    int commaCount = 1;  
    string item = "";  
    for(size_t x = 0; x < record.length(); x++)  
    {  
        if (record[x] == ',')  
        {  
            commaCount = commaCount + 1;  
            continue;  
        }  
        else if (commaCount == field)  
        {  
            item = item + record[x];  
        }  
    }  
    return item;  
}  
  
void readData(string names[], string passwords[], int &indx)  
{  
    indx = 0;  
    ifstream file("UAMS(Pass).txt");  
    if(!file.is_open())
```

```
{
    return;
}

string record;
while(getline (file, record))
{
    names[indx] = getfield (record, 1);
    passwords[indx] = getfield(record, 2);
    indx++;

    if(indx >= 100) break;
}
file.close();
}

int main()
{
    system("cls");
    int ch;
    bool loggedin = false;

    while (!loggedin){
        system("cls");
        cout<<" #=====#"<<endl;
        cout<<" # University Admission Management System #"<<endl;
```

```
    cout<<"
#=====#\n"<<endl;

    cout<< "1. Register User"<< endl;
    cout<< "2. Login"<< endl;

    cout<< "Choose option: ";
    cin>>ch;

    if (ch == 1){
        Register();
    }
    else if (ch == 2){
        if (login()){
            cout<< "\033[32mLogin successfull welcome!\033[0m " <<endl;
            loggedin = true;
            cout<< "Press Any Key To Continue To Menu...." << endl;
            getch();
            break;
        }
        else {
            cout<< "\033[31mInvalid username or password! (Access
Denied....)\033[0m" << endl;
            cout<< "Press Any Key To Try Again....."<< endl;
            getch();
        }
    }
```

```
}  
else {  
    cout<< "Invalid Option."<< endl;  
    getch();  
}  
}
```

```
string name[100], Cnic[100];  
float Matric[100], Fsc[100], ECAT[100], Merit[100];  
int indx = 0;  
indx = loadFromFile(name, Cnic, Matric, Fsc, ECAT, Merit);
```

```
int op;  
int num;
```

```
while(true){  
    Header ();  
    Menu();  
    cout<<"\033[32mCHOOSE AN OPTION FROM MENU\033[0m\n:";  
    cin>>op;
```

```
if (op == 1) {
```

```
    cout<< "How many students you want to Add: ";
```

```
    cin>> num;
    indx = addstudent(name , Cnic , Matric , Fsc , ECAT , Merit , num , indx);
    saveToFile(name, Cnic, Matric, Fsc, ECAT, Merit, indx);
    cout<< "\n\033[32mStudents Added Successfully!\033[0m\n";

}

else if (op == 2){

    cout<<"You Have Entered Option:2"<<endl;
    update(name , Cnic , Matric , Fsc , ECAT , Merit , indx);
    saveToFile(name, Cnic, Matric, Fsc, ECAT, Merit, indx);
}

else if(op == 3){

    cout<<"You Have Entered Option:3"<<endl;
    merit(name , Cnic , Matric , Fsc , ECAT , Merit ,indx);
    saveToFile(name, Cnic, Matric, Fsc, ECAT, Merit, indx);

}

else if(op == 4){
```

```
cout<<"You have entered option :4"<< endl;
Delete(name, Cnic, Matric, Fsc, ECAT, Merit , indx);
saveToFile(name, Cnic, Matric, Fsc, ECAT, Merit, indx);

}

else if(op == 5){

    cout<<"You Have Entered Option:5"<<endl;
    View(name , Cnic , Matric , Fsc , ECAT , Merit , indx);

}

else if(op == 6){
    cout<< "\033[33mExiting Program.....\033[30m\n";
    return 0;
}
else
{
    cout<< "Invalid Option. Try again.\n";
}
cout<<"Press Any Key To Continue.....";
getch();
system("cls");
```

```
}  
  
}  
  
int loadFromFile(string name[], string Cnic[], float Matric[], float Fsc[], float  
ECAT[], float Merit[])  
{  
    ifstream fin("Student Credentials.txt");  
  
    int i = 0;  
  
    if(!fin) return 0;  
  
    while(getline(fin,name[i], '|') && getline(fin, Cnic[i], '|') && fin >> Matric[i]  
&& fin.ignore() && fin>> Fsc[i] && fin.ignore() && fin >> ECAT[i] &&  
fin.ignore() && fin >> Merit[i])  
    {  
        fin.ignore();  
        i++;  
    }  
    return i;  
}  
  
void saveToFile(string name[], string Cnic[], float Matric[], float Fsc[], float  
ECAT[], float Merit[], int indx)  
{  
    ofstream fout("Student Credentials.txt");  
  
    for(int i = 0; i < indx; i++){
```

```
        fout << name[i] << "|" << Cnic[i] << "|" << Matric[i] << "|" << Fsc[i] << "|"
<< ECAT[i] << "|" << Merit[i] << endl;
```

```
    }
```

```
    }
```

```
void Register()
```

```
{
```

```
    system("cls");
```

```
    cout << " |=====|"
<<endl;
```

```
    cout<< " |<<<<<<<<<<<< USER Registration >>>>>>>>>>>>|" <<endl;
```

```
    cout << " |=====|\n"
<<endl;
```

```
string user,pass;
```

```
cout<<"Enter new username: ";
```

```
cin>>user;
```

```
cout<<"Enter new password:";
```

```
cin>>pass;
```

```
ofstream file("UAMS(Pass).txt" , ios::app);
```

```
if(!file.is_open()){
```

```
    cout<< " Error opening UAMS(Pass).txt for writing " <<endl;
```

```
    return;
```

```
}
```

```
file << user << "," << pass << endl;
```

```
file.close();

cout<< "\033[32mUser Registered Successfully!\033[0m" <<endl;
cout<<"Press any key to continue login..." <<endl;
getch();

}

bool login()
{
    system("cls");
    cout<<"#*****#
"<<endl;

    cout<<"#===== UAMS LOGIN
=====#"<<endl;

    cout<<"#*****#\n
"<<endl;

    string username, password,user, pass;
    cout<< "Enter Username: ";
    cin>> username;
    cout<< "Enter Password: ";
    cin>>password;

    ifstream file("UAMS(Pass).txt");
    if(!file.is_open()){
        return false;
```

```
    }
    string record;
    while (getline(file, record))
    {
        user = getfield(record, 1);
        pass = getfield(record, 2);

        if(user == username && pass == password){
            file.close();
            return true;
        }
    }
    file.close();
    return false;
}

int isPresent(string name, string password, string names[], string passwords[], int
idx)
{
    int index = -1;
    for(int x = 0; x < idx; idx++)
    {
        if(name == names[x] && password == passwords[x])
            index = x;
    }
    return index;
}
```

```
}

void Header() {
    system("cls");
    cout<<"#=====#"<<endl;
    cout<<"# University Admission Management System #"<<endl;
    cout<<"#=====#\n"<<endl;
}

void Menu() {

    cout<<"    #=====# "<<endl;
    cout<<"    ##<<<<<<  MENU >>>>>>## "<<endl;
    cout<<"    #=====# \n"<<endl;


    cout<<"1 - Add Students."<<endl;
    cout<<"2 - Update record."<<endl;
    cout<<"3 - Calculate Merit."<<endl;
    cout<<"4 - Delete Student."<<endl;
    cout<<"5 - View."<<endl;
    cout<<"6 - Exit. "<<endl;
}

int addstudent(string name[],string Cnic[],float Matric[],float Fsc[],float ECAT[] ,
float Merit[], int num,int indx) {

    cout<<"*===== ADDING STUDENT
=====*\n"<<endl;

    ofstream fout("Student Credentials.txt" , ios::app);
    if(!fout) {
```

```
    cout<<" Error opening file!" <<endl;
    return indx;
}
cin.ignore();

for(int idx = indx; idx < indx + num; idx++){
    cout<<" \nEnter the Name: ";
    getline(cin,name[idx]);
    cout<<"Enter the CNIC:";
    cin>>Cnic[idx];
    cout<<"Enter the Matric Marks (Out of 1100): ";
    cin>>Matric[idx];
    cout<<"Enter the FSC Marks (out of 1200): ";
    cin>>Fsc[idx];
    cout<<"Enter the ECAT Marks:";
    cin>>ECAT[idx];
    cin.ignore();
    fout << name[idx] << "," << Cnic[idx] << "," << Matric[idx] << "," << Fsc[idx]
    << "," << ECAT[idx] << endl;
}

    fout.close();

    return indx + num;
}

int update(string name[],string Cnic[],float Matric[],float Fsc[],float ECAT[] ,
float Merit[], int indx) {
```

```
cout<<"*===== UPDATING STUDENT
=====*\n          "<<endl;

if (indx == 0) {
    cout<< "NO records available to update .\n";
    return indx;
}

string searchCnic;
int i = -1;
View(name, Cnic , Matric , Fsc , ECAT , Merit , indx);
cout << "Enter Student CNIC to update:\n ";
cin >> searchCnic;

for(int j = 0; j < indx; j++){
    if(Cnic[j] == searchCnic){
        i = j;
        break;
    }
}

if(i >= 0 && i < indx) {
    cin.ignore();
    cout<< "Enter New Student Name: ";
    getline(cin , name[i]);
    cout<< "Enter New CNIC: ";
    cin>>Cnic[i];
    cout<< "Enter New Matric marks:";
    cin >> Matric[i];
```

```
    cout << "Enter New FSC Marks: ";
    cin >> Fsc[i];
    cout << "Enter New ECAT Marks: ";
    cin >> ECAT[i];
    cin.ignore();

    cout<< "Student Updated Successfully.\n";
    return indx;
}
else {
    cout << "Student Not Found. \n";
    return indx;
}
}

void merit(string name[], string Cnic[], float Matric[], float Fsc[], float ECAT[],
float Merit[], int indx){
    if (indx == 0){
        cout << "No students to calculate merit for.\n";
        return;
    }
    for(int i = 0; i < indx; i++){
        float a =(Matric[i]/1100)*100;
        float b =(Fsc[i]/1200)*100;
        float c =(ECAT[i]/400)*100;
        Merit[i] = (a * 0.10) + (b * 0.40) + (c * 0.50);
```

```
}  
cout<< "Calculated Merit: \n";  
for (int i = 0; i < indx; i++){  
    cout<< name[i] << " -> Merit : " << Merit[i] << endl;  
}  
}  
  
void Delete(string name[],string Cnic[],float Matric[],float Fsc[],float ECAT[] ,  
float Merit[], int &indx) {  
    cout<<"  |*Delete Student*|\n  " <<endl;  
  
    if (indx == 0) {  
        cout<< "NO records available to Delete .\n";  
        return;  
    }  
    int i;  
    View(name, Cnic , Matric , Fsc , ECAT , Merit , indx);  
    cout << "Enter student number to Delete: ";  
    cin >> i;  
    i--;  
  
    if(i >= 0 && i < indx) {  
        cin.ignore();  
        for( int a = i; a < indx - 1; a++){  
            name[a] = name[a + 1];  
            Cnic[a] = Cnic[a + 1];  
            Matric[a] = Matric[a + 1];
```

```
Fsc[a] = Fsc[a + 1];
ECAT[a] = ECAT[a + 1];
Merit[a] = Merit[a + 1];
} indx--;
cout<< "\033[32mStudent deleted Successfully.\033[0m\n";
}
else {
    cout << "\033[31mInvalid Student number.\033[0m \n";
}
}

void View(string name[], string Cnic[], float Matric[], float Fsc[], float ECAT[],
float Merit[], int &indx){
    if (indx == 0) {
        cout<< "No students record to view.\n";
        return;
    }

    cout<< "      \033[33m*<<<<===== STUDENTS RECORD
=====>>>*\n\033[0m      "<<endl;

    cout << left << setw(5) << "No." << left << setw(25) << "Name" << left <<
    setw(25) << "CNIC" << left << setw(15) <<"Matric" << left << setw(15) <<
    "FSC" << left << setw(15) << "ECAT" << left << setw(15) << "Merit"
    <<endl; for(int i = 0; i < indx; i++){

        cout << left << setw(5) << i + 1 << left << setw(25) << name[i] << left <<
        setw(25) << Cnic[i] << left << setw(15) << Matric[i] << left << setw(15) << Fsc[i]
        << left << setw(15) << ECAT[i] << left << setw(15) << Merit[i] <<endl;
    }
}
```

Weakness In Application

- The system works only in console mode.
- The program supports a fixed number of students.
- There is no error handling for invalid input values.
- The interface is not user-friendly as the graphics are not professional.

Future Direction:

- A graphical user interface (GUI) can be added.
- A database can be used instead of text files.
- Input validation and security can be improved.
- Search and filter options can be enhanced.
- The system can be converted into a web-based application.